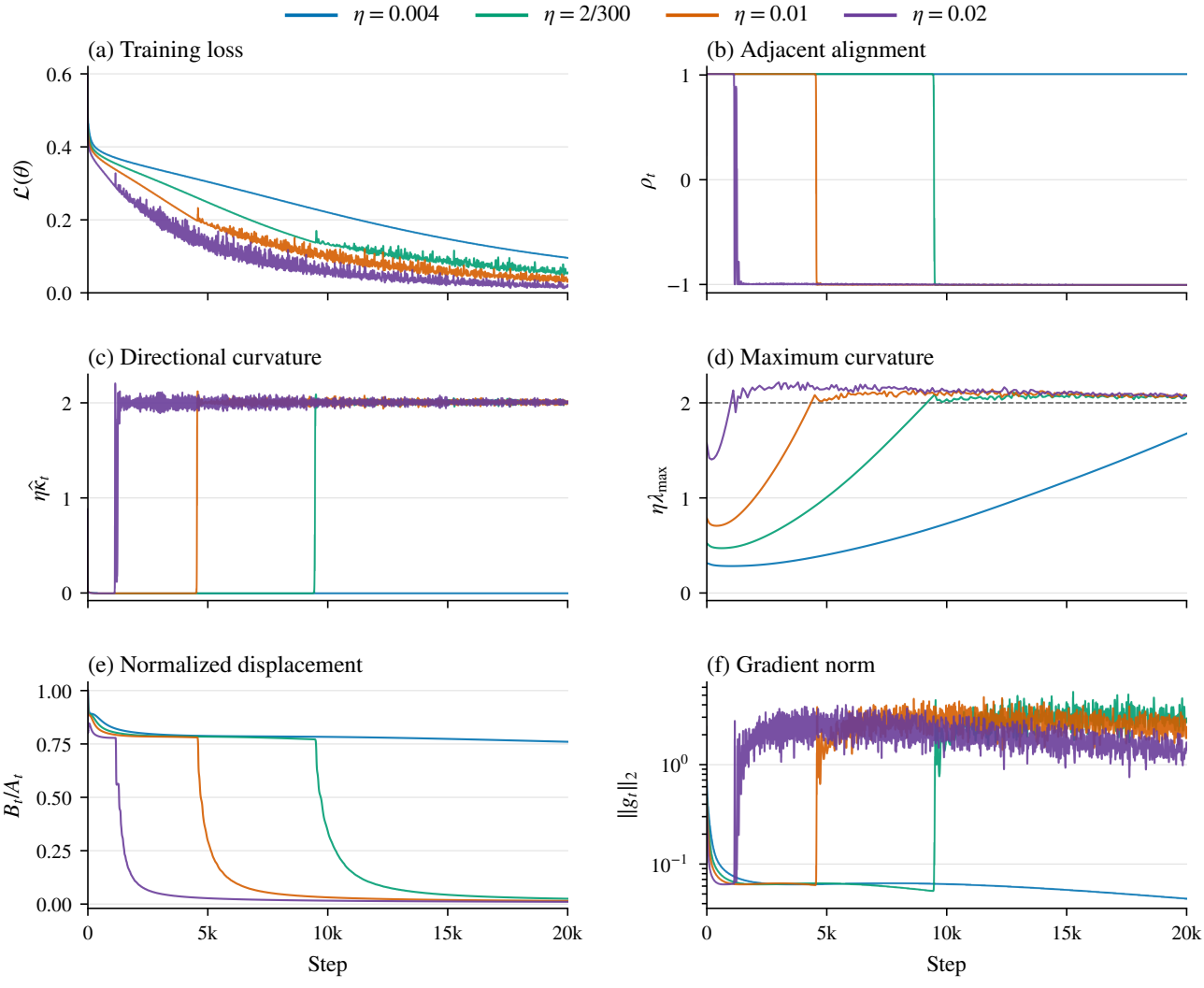

$\eta = 0.004$
$\eta = 2/300$
$\eta = 0.01$
$\eta = 0.02$
(a) Training loss
$\mathcal{L}(\theta)$
0.6
0.4
0.2
0.0
(b) Adjacent alignment
ρ_t
1
0
−1
(c) Directional curvature
$\eta\hat{\kappa}_t$
2
1
0
(d) Maximum curvature
$\eta\lambda_{\max}$
2
1
0
(e) Normalized displacement
B_t/A_t
1.00
0.75
0.50
0.25
0.00
0
5k
10k
15k
20k
Step
(f) Gradient norm
$\|g_t\|_2$
10^0
10^{-1}
0
5k
10k
15k
20k
Step